# Role of Vitamin-D and Its Deficiency Associated Jeopardy in the Athletic Population

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## ABSTRACT

*Vitamin D is a fat-soluble vitamin, known to play an important role in maintaining bone health and calcium homeostasis. According to the emerging literature, beyond this established role, vitamin D also plays a central role in muscle function, gene signaling, protein synthesis, hormone synthesis, cell growth, and inflammatory response, etc. Because of its diverse functions vitamin D gained the attention of many researchers in recent years. Vitamin D deficiency is very prevalent in many countries worldwide including India and its deficiency leads to rickets in children and osteomalacia in adults. In addition to osteomalacia, vitamin D deficiency may also cause performance detriment in athletes, by decreasing the contraction force and by increasing muscle atrophy of fast-twitch fibers. This review addresses the role of vitamin D in maintaining athletic performance and vitamin D deficiency-related risks in the athletic population.*

**Keywords:**

Rickets, osteomalacia, atrophy, homeostasis, gene signaling

## INTRODUCTION

Vitamin D also called as cholecarciferol is a fat-soluble vitamin. Two vitamin D pro-vitamins have been found in nature. Ergosterol (vitamin D2) is found in plants and cholecalciferol (Vitamin D3) is normally produced in the skin by UVB irradiation of 7-dehydrocholesterol. Cutaneous production of vitamin D is slightly influenced by altitude, latitude, pollution, season, skin color and age. Vitamin D produced at the epidermis enters the circulation and undergo two hydroxylation reactions in the liver and in kidneys and produces 25- hydroxyl vitamin D (25(OH)D) and 1,25-dihydroxy vitamin D (1,25(OH)2D) successively. 1,25(OH)2D also called calcitriol (contains 3 hydroxy groups) is the biologically active form of vitamin D. Calcitriol acts as a steroid hormone while regulating the blood calcium levels. Primarily it maintains the blood calcium levels by increasing calcium uptake from the intestine and by decreasing the calcium excretion through kidneys. Calcitriol mediates its action by binding to a nuclear receptor called vitamin D receptor (VDR). Vitamin D acting through VDR regulates calcium absorption in the intestine by enhancing the expression of transient

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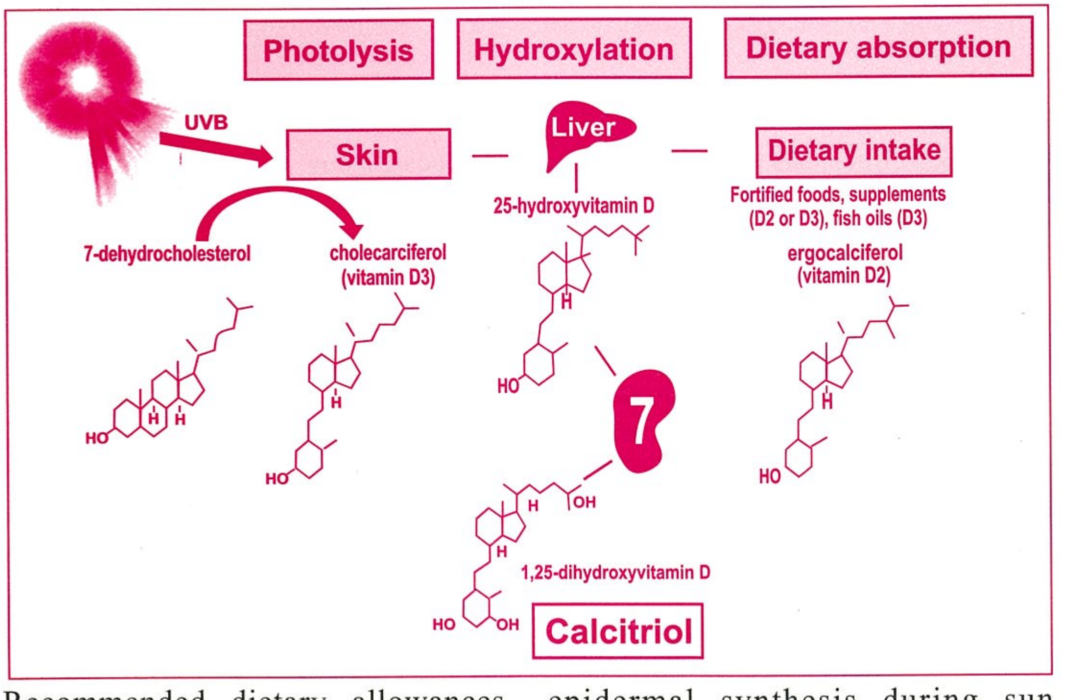
receptor potential cation channel subfamily V member 6 (TRPV6- that transports calcium from lumen to cell) gene and calbindin 9k (transports calcium athwart the enterocytes) gene. In earlier times Vitamin D was anticipated to be important for bone growth, density, and osteoclast activity (Larson-Meyer & Willis, 2010) but, emerging literature infer the vitamin D's central role in muscle function, gene signaling, inflammatory response, protein synthesis, hormone synthesis, minimizing infection risk and cell growth, etc (Campbell & Allain, 2006; Ceglia, 2008; Larson-Meyer, Willis, Smith, & Broughton, 2012). Besides epidermal synthesis through photolysis from 7-dehydro cholesterol, some amount of Vitamin D (20%) can also be obtained through diet. Some common foods which contain significant levels of vitamin D are salmon, fatty fish, egg yolks, cheese, orange juice (Chen et al., 2007). Fish liver oil is the richest source of vitamin D. Vitamin D2 can be

**Figure 1: Synthesis of cholecalciferol and calcitriol**




Recommended dietary allowances (RDA) for Vitamin D: The RDA is the average daily intake sufficient to meet the nutrient necessities of nearly all healthy individuals. Though we obtain a major amount of vitamin D through epidermal synthesis during sun exposure, vitamin D RDAs are set on the basis of minimal sun exposure. As per the vitamin requirement guidelines established by the Institute of Medicine (IOM), the estimated average requirement (EAR) and recommended dietary allowance (RDA) of Vitamin D for adult males and females are set as 400 IU (10µg) and 600IU (15µg) respectively (Ross AC et al., 2012). However, for elderly people (above70 years), Vitamin D RDA is set as 800 IU (20µg). On the other hand, according to the recommendations given by Endocrine Society for people at risk for vitamin D deficiency, vitamin D RDA is set as 400–1000 IU/day for infants, 600–1000 IU/day in children (1–18 years) and 1500–2000IU in adults. However, to raise the blood level of 25(OH)D consistently above 30 ng/ml at least 1000 IU/d of vitamin D is recommended (Holick et al., 2011).

What is considered a deficiency?

Two sets of reference ranges are framed for vitamin D. One is by the Institute of Medicine (IOM) and the other is by the Endocrine Society. The Institute of Medicine (IOM) defined vitamin D deficiency as blood 25(OH) D levels less than 20 ng/mL (Calcium, Ross, Taylor, Yaktine, & Valle, 2011; Holick et al., 2011). However, after considering the other additional benefits of vitamin D in improving muscle function and reducing several chronic illnesses, the Endocrine Society referred the 25(OH)D levels between 21–29 ng/mL as vitamin D insufficiency (Holick et al., 2011). Furthermore many studies have suggested that, when 25(OH)D level is between 30 - 40 ng/mL, PTH levels decrease and reach the plateau indicating the maximum effect of vitamin D on calcium metabolism and bone health (Holick, 2011; Holick et al., 2005). Gagnon et al. conducted a study among 4,164 adults (mean age 50 years) to evaluate the prospective association between Vitamin D levels and Metabolic Syndrome (MeS). After 5 years of follow-up they found out that the adults with blood 25(OH)D level of <18 ng/mL and 18–23 ng/mL had an increased risk for developing metabolic syndrome compared to adults who had a blood 25(OH) D level of >34 ng/mL (Gagnon et al., 2012). The same was also validated in other studies (Deleskog et al., 2012).

| | IOM Guidelines | | Endocrine Society Guidelines | |
|---|---|---|---|---|
| | ng/mL | nmol/L | ng/mL | nmol/L |
| Deficient | < 12 | < 30 | < 20 | < 50 |
| Insufficient | 12 to 20 | 30-50 | 21-29 | 52.5-72.5 |
| Sufficient | > 20 | > 50 | 30-60 | 75-150 |
| Ideal | | | 40-60 | 100-150 |
| Considered safe | | | < 100 | < 250 |
| Toxic | | | > 150 | > 375 |

**How to diagnose vitamin D deficiency?**

Vitamin D deficiency can be diagnosed by doing the blood test. For this, two components of vitamin D can be measured in the blood. One is 1,25(OH)2D and the other is 25(OH)D. However, 1,25(OH)2D assessment is not a good measure to monitor the vitamin D status because of its short half-life (6-8 hours) and blood 1,25(OH)2D status also fluctuates with blood parathyroid hormone (PTH), calcium, and phosphorus levels. So 25(OH)D estimation is the most accurate way to measure the vitamin D status as it reflects the sum of vitamin D derived from diet, supplements as well as epidermal synthesis. Apart from 25(OH)D, estimation of other markers like PTH and alkaline phosphatase (ALP) may provide some additional information. Blood PTH concentration increases when blood vitamin D concentration decreases below 25-50 nmol/L (Holick, 2007). ALP estimation is helpful to distinguish bone damage from vitamin D deficiency (osteomalacia) from general low bone mineral density (BMD) or osteoporotic bone.

**Prevalence of Vitamin D deficiency:**

Vitamin D deficiency is one of the widespread deficiencies in India. Based on the reports obtained from different community-based studies, the prevalence of vitamin D deficiency among the healthy controls from the general population of India may range from 50% to 94% (Aparna, Muthathal, Nongkynrih, & Gupta, 2018). Still, the athletic population is yet to be fully examined.

**Factors causing vitamin D deficiency in athletes:**

Though most of the athletes spend seemingly adequate time under the sun, they avoid training during peak sunlight hours by opting practice early in the morning or late in the evening. So vitamin D deficiency is very common in athletes. Too much time spent indoors, living at high altitudes plus poor nutrition might result in vitamin D deficiency in many athletes. Apart from these, some other factors like latitude, pollution, skin color, gut health, outfits and usage of sunscreen lotions (SPF-sun protection factor) also decide the occurrence of vitamin D deficiency. Melanin reduces the skin's capacity to produce vitamin D3 by absorbing the UVB radiation from the sun. Therefore dark-skinned athletes require longer hours of sun exposures to make the same amount of vitamin D as compared to a person with a white skin tone. Disorders that affect the gut, such as Crohn's disease, can also undermine the intestines' ability to absorb vitamin D. Vitamin D deficiency is likely if athletes follows a strict vegan diet, since most of the natural sources are animal-based, including fish and fish oils, egg yolks, fortified milk, and beef liver. Consumption of foods rich in vitamin D and foods that have been fortified with vitamin D reduces the risk of vitamin D deficiency. Various studies have been conducted across the world to understand the prevalence and causative of vitamin D deficiency among the athletic population. Based on those reports it is found that the vitamin D deficiency is highly prevailing in athletes. Lack of sun exposure (Indoor training) Season (winter), Latitude appear to be the major factors which results in vitamin D deficiency in athletic population (Cannell et al., 2006; G. L. Close et al., 2013; Graeme L. Close et al., 2013; Constantini, Arieli, Chodick, & Dubnov-Raz, 2010; Flueck, Schlaepfer, & Perret, 2016; Halliday et al., 2011, 2011; Hamilton, Grantham, Racinais, & Chalabi, 2010; Lappe et al., 2008; Larson-Meyer & Willis, n.d.; Larson-Meyer et al., 2012; Lovell, 2008).

**Vitamin D and athletic performances**

Many studies have examined the impact of vitamin D supplementation on athletic performance. Better performance observed during the summer months than winter months in many studies is accredited to the optimum vitamin D produced due to increased exposure to ultraviolet (UVB) light during summer months. Barker T et al in 2013 recognized that the vitamin D supplementation (4,000 IU for 28 days) in moderately active adults not only increased the serum 25(OH)D concentrations and also enhanced the recovery of peak isometric force after a physically stressful event (Barker,T., Schneinder,E.D., Dixon, B. M., Henriksen, V.T., & Weaver,L.K.,2013). Wyon et al. studied the effect of vitamin D supplementation on muscle fitness parameters and injury prevalence in elite ballet dancers. They divided 24 elite classical ballet dancers into intervention (17 ballet dancers) and control groups (7 ballet dancers). 2000 IU of vitamin D3 oral supplementation was given to the intervention group for 4 months and control groups were given placebo for the same length of time. Injury rates are noted throughout the period and isometric muscular strength & vertical jump heights are measured at baseline and at the end of the study. Wyon et al reported, less incidence of injuries besides the improved isometric strength and vertical jump heights in the supplemented group when compared to the control group (Wyon, Koutedakis, Wolman, Nevill, & Allen, 2014). Close et al, conducted a study to examine the effect of vitamin D supplementation on musculoskeletal performance of professional athletes who lived at the northerly latitude (UK-53oN). In this study, athletes were supplemented with 5,000 IU/day of vitamin D3 for 8 weeks and the placebo was given to the control group. They reported significant improvements in both 10-m sprint times and vertical jump heights along with the improved blood levels of 25(OH)D in the test group. However, no changes have been observed in the blood 25(OH)D levels and physical performance in the control group (G. L. Close et al., 2013). On the basis of available review of the literature on PubMed (between 1930s-2012), Shuler et al (Shuler, Wingate, Moore, & Giangarra, 2012) reported a strong correlation between sufficient blood levels of 25(OH)D (>30 ng/mL) and musculoskeletal performance parameters such as jump height, jump velocity, jump power, exercise capacity, and physical performance. Shuler et al also reported that the muscle protein synthesis and adenosine triphosphate levels enhances with an increase in the levels of 25(OH)D (between 30-50 ng/mL) in blood and optimum 25(OH) D levels are also associated with reduced inflammation, pain, and myopathy and prevention of fractures in athletes. Ward K.A et al (2009) also noticed a positive correlation of Vitamin D levels with muscle power, force, jump velocity, jump height, and Esslinger Fitness Index in post-menarchal adolescent girls (Ward et al., 2009).

**Vitamin D deficiency and associated jeopardy:**

Several researchers provided evidence to support the importance of optimum blood vitamin D levels in maintaining bone health and in the prevention of bone and muscle injuries. Ruohola et al., in 2006 have revealed that the instances of stress fracture were 3.6 times higher in Finnish military recruits with lower blood vitamin D concentrations (<30ng/ml or <75 nmol/L)(Ruohola et al., 2006). On the other hand; the inclusion of vitamin D (800 IU/day) supplementation in combination with calcium (2000mg) reduced the incidence of stress fractures by 21% in female U.S. Navy recruits (Lappe et al., 2008; Ruohola et al., 2006). Sato et al conducted a study on elderly women with post-stroke hemiplegia in 2005 to understand the effect of vitamin D supplementation in prevention of falls and hip fractures and noticed increased type II muscle fibers number and size and decreased incidence of falls (59%) after vitamin D supplementation and concluded that the Vitamin D may increase muscle strength by improving atrophy of type II muscle fibers (Sato, Iwamoto, Kanoko, & Satoh, 2005). After reviewing many animal and tissue culture studies, Girgis et al (2013) reported that vitamin D deficiency is associated with atrophy of type II (fast-twitch) muscle fibers and myalgia (muscle pain). Vitamin D deficiency also prolongs the time to peak contractile tension and relaxation by impairing calcium uptake in the sarcoplasmic reticulum (Girgis, Clifton-Bligh, Hamrick, Holick, & Gunton, 2013). Besides declining the muscle strength and increasing the muscle atrophy vitamin D insufficiency also seems to delay rehabilitation and recovery of the muscle after orthopedic surgery (Barker et al., 2011; Kiebzak, Moore, Margolis, Hollis, & Kevorkian, 2007). Therefore upholding optimum vitamin D status is not only beneficial for athletes to maintain their health but also to peak up their performance and to become less susceptible to injuries.

**Role of vitamin D in immunity and inflammation:**

Vitamin D is one of the influential regulators of inflammation and innate immunity. Cells of the innate immune system, especially monocytes, macrophages, natural killer cells and epithelial cells of the respiratory tract helps to defend against invading pathogenic bacteria, fungi, and viruses by secreting the antimicrobial peptides(AMP) (Gombart, Borregaard, & Koeffler, 2005). Since vitamin D has the ability to turn on the gene expression of broad-spectrum antimicrobial peptides (AMP), the vulnerability to influenza and common cold is markedly influenced by the seasonal fluctuations of vitamin D levels (Cannell, Hollis, Zasloff, & Heaney, 2008; Cannell et al., 2006; Larson-Meyer & Willis, 2010). Upper respiratory tract infection (URTI) is a prevailing health issue, especially noticed in the athletic population. In athletes, prolonged intense training subdues the innate immune function and enhances the incidence of upper respiratory tract infections (URTIs). Recent studies found that the frequency and severity of URTIs are more in athletes with vitamin D insufficiency compared to those with optimum vitamin D levels (Halliday et al., 2011). Besides regulating the immunity vitamin D also control inflammation by increasing the production of various anti-inflammatory cytokines (Transforming growth factor-TGF and interleukins namely IL-4, IL-10, and IL-13) and by reducing the production of pro-inflammatory cytokines IL-6, IL-2, interferon-γ, and tumor necrosis factor-alpha (TNFα) (Barker et al., 2014; Larson-Meyer et al., 2012).

**Vitamin D toxicity:**

Vitamin D toxicity has been of great concern, especially for children. However, as per the IOM guidelines, vitamin D supplementation of up to 4,000 IUs/day is recommended as safe for most children and adults. A study done by Heaney et al depicted that the Vitamin D3 supplementation of 10,000 IUs/day for 5 months did not cause any untoward toxicity in healthy adult males (Heaney, Davies, Chen, Holick, & Janet Barger-Lux, 2003). The IOM and the Endocrine Society also recognized that patients with kidney stones or with primary hyperparathyroidism can receive vitamin D supplementation without concern of mounting risk of hypercalcemia and kidney stones. On the other hand toxicity from sunlight or artificial UVB exposure is not possible because of the negative feedback mechanisms that operate during the prolonged sun exposure (Institute of Medicine (US) Committee to Review Dietary Reference Intakes for Vitamin D and Calcium.,2011).

**Summary:** Vitamin D plays a major role in upholding the musculoskeletal strength and functions thus can influence physical performance. So athletes who are engaged in indoor training, those who follow restricted diets need to be screened for vitamin D defficiency. Furthermore, the athletes who exhibit the incidence of stress fractures, musculoskeletal pain, bone, joint injury, and frequent sickness (URTIs) or signs of overtraining should be evaluated for vitamin D deficiency. If an athlete is found to have a vitamin D deficiency, appropriate supplementation should be prescribed by a physician or registered dietitian.

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